

OSSP (Operating Systems and Systems Programming) – 25CS2104E

OSSP LAB – SKILL-1

Aim

To set up a Linux development environment by installing a Linux VM, configuring GCC, creating a Git repository, organizing the project structure, understanding shell architecture, and building an initial Makefile, and to analyze process abstraction by executing `fork()`, understanding the `exec()` family, studying parent-child relationships, inspecting the process tree, and practicing system-call tracing.

Algorithm

1. Install and configure the Linux virtual machine.
2. Install and verify the GCC compiler.
3. Create a project directory with an appropriate structure.
4. Initialize a Git repository and track the project files.
5. Study the basic architecture and working of the Linux shell.
6. Create an initial Makefile to compile and manage the C programs.
7. Develop a C program using `fork()` to create parent and child processes.
8. Display the PID and PPID of the parent and child processes.
9. Study the `exec()` family of system calls and observe how a process executes a new program.
10. Analyze the relationship between parent and child processes.
11. Inspect the process hierarchy using Linux commands such as `ps` and `pstree`.
12. Use `strace` to trace and analyze the system calls made by the programs.
13. Record and analyze the observations.
14. Stop the experiment.

Step 1

```
bhavana@Bhavana:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [446 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.6 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [111 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [150 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [84.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [235 kB]
```

Step 2

```
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3052 B]
Fetched 2873 kB in 6s (448 kB/s)
45 packages can be upgraded. Run 'apt list --upgradable' to see them.
bhavana@Bhavana:~$ sudo apt upgrade
Upgrading:
  apparmor                iproute2                libsystemd-shared       systemd
  apport                  libapparmor1            libsystemd0              systemd-cryptsetup
  apport-core-dump-handler libatk-bridge2.0-0t64  libudev1                 systemd-resolved
  at-spi2-common          libatk1.0-0t64         linux-libc-dev           systemd-sysv
  at-spi2-core            libatspi2.0-0t64       motd-news-config         tzdata
  base-files              libgcrypt20             openssl                  ubuntu-minimal
  cloud-init              libnss-systemd          openssl-provider-legacy ubuntu-release-upgrader-c
  cloud-init-base         libpackagekit-glib2-18 packagekit                ubuntu-wsl
  distro-info-data        libpam-systemd          python3-apport           udev
  gir1.2-packagekitglib-1.0 libpciaccess0           python3-distupgrade      wsl-setup
  gtk-update-icon-cache   libssl3t64             python3-problem-report

Not upgrading yet due to phasing:
  python3-software-properties  software-properties-common

Summary:
  Upgrading: 43, Installing: 0, Removing: 0, Not Upgrading: 2
  6 standard LTS security updates
  Download size: 18.4 MB
  Space needed: 20.5 kB / 1024 GB available

Continue? [Y/n] y
```

```

Continued from y
Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 motd-news-config all 14ubuntu6.1 [5632 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.1 [75.0 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsystemd0 amd64 259.5-0ubuntu3.3 [480 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-sysv amd64 259.5-0ubuntu3.3 [9252 B]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-resolved amd64 259.5-0ubuntu3.3 [298 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpam-systemd amd64 259.5-0ubuntu3.3 [274 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsystemd-shared amd64 259.5-0ubuntu3.3 [2462 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd amd64 259.5-0ubuntu3.3 [3129 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 systemd-cryptsetup amd64 259.5-0ubuntu3.3 [124 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 udev amd64 259.5-0ubuntu3.3 [1501 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libudev1 amd64 259.5-0ubuntu3.3 [150 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssl-provider-legacy amd64 3.5.5-1ubuntu3.3 [39.7 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libssl3t64 amd64 3.5.5-1ubuntu3.3 [2361 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgcrypt20 amd64 1.12.0-2ubuntu1 [671 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 distro-info-data all 0.72-0ubuntu0.26.04.1 [6118 B]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1.1 [1198 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libapparmor1 amd64 5.0.2-0ubuntu1-26.04.1 [51.3 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnss-systemd amd64 259.5-0ubuntu3.3 [226 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssl amd64 3.5.5-1ubuntu3.3 [1243 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 tzdata all 2026c-0ubuntu0.26.04.1 [193 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-minimal amd64 1.570.2 [8594 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apparmor amd64 5.0.2-0ubuntu1-26.04.1 [380 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.22 [24.9 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrader all 1:26.04.22 [102 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 2.34.1-0ubuntu0.1 [17.4 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apport all 2.34.1-0ubuntu0.1 [96.4 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu0.1 [9674 B]
Get:28 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport all 2.34.1-0ubuntu0.1 [83.9 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 at-spi2-common all 2.60.4-0ubuntu0.1 [6604 B]
Get:30 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatk-bridge2.0-0t64 amd64 2.60.4-0ubuntu0.1 [67.2 kB]
Get:31 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatk1.0-0t64 amd64 2.60.4-0ubuntu0.1 [53.5 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 at-spi2-core amd64 2.60.4-0ubuntu0.1 [59.6 kB]
Get:33 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatspi2.0-0t64 amd64 2.60.4-0ubuntu0.1 [85.5 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu3-26.04.1 [587 kB]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu3-26.04.1 [2110 B]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpackagekit-glib2-18 amd64 1.3.4-3ubuntu1.1 [121 kB]
Get:37 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-packagekitglib-1.0 amd64 1.3.4-3ubuntu1.1 [25.5 kB]
Get:38 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 gtk-update-icon-cache amd64 4.22.4+ds-0ubuntu0.1 [42.7 kB]
Get:39 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpciaccess0 amd64 0.18.1-1ubuntu4.1 [19.0 kB]
Get:40 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-libc-dev amd64 7.0.0-29.29 [1619 kB]
Get:41 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 wsl-setup amd64 0.6.3ubuntu-26.04.1 [10.3 kB]
Get:42 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-wsl amd64 1.570.2 [8518 B]
Get:43 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 packagekit amd64 1.3.4-3ubuntu1.1 [505 kB]
Fetched 18.4 MB in 22s (835 kB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database ... 42255 files and directories currently installed.)
Preparing to unpack .../motd-news-config_14ubuntu6.1_all.deb ...
Unpacking motd-news-config (14ubuntu6.1) over (14ubuntu6) ...
Preparing to unpack .../base-files_14ubuntu6.1_amd64.deb ...
Unpacking base-files (14ubuntu6.1) over (14ubuntu6) ...
Setting up base-files (14ubuntu6.1) ...
motd-news.service is a disabled or a static unit not running, not starting it.
(Reading database ... 42255 files and directories currently installed.)
Preparing to unpack .../libsystemd0_259.5-0ubuntu3.3_amd64.deb ...
Unpacking libsystemd0:amd64 (259.5-0ubuntu3.3) over (259.5-0ubuntu3) ...
Setting up libsystemd0:amd64 (259.5-0ubuntu3.3) ...
(Reading database ... 42255 files and directories currently installed.)
Preparing to unpack .../systemd-sysv_259.5-0ubuntu3.3_amd64.deb ...

```

Step 3

```

Processing triggers for libcolord icon theme (0.16-2ubuntu1) ...
bhavana@Bhavana:~$ sudo apt install gcc
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2

```

Step 4

```

bhavana@Bhavana:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

Step 5

```
bhavana@Bhavana:~$ sudo apt install git
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
bhavana@Bhavana:~$ git --version
```

Step 6

```
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
bhavana@Bhavana:~$ git --version
git version 2.53.0
```

Step 7

```
bhavana@Bhavana:~$ git config --global user.name "Bhavana"
bhavana@Bhavana:~$ git config --global user.name "bhavana"
bhavana@Bhavana:~$ git config --global user.email "kunamanenibhavana@gmail.com"
bhavana@Bhavana:~$ git config --global --list
user.name=bhavana
user.email=kunamanenibhavana@gmail.com
bhavana@Bhavana:~$
```

Step 8

```
user.email=kunamanenibhavana@gmail.com
bhavana@Bhavana:~$ cd ~
```

```
bhavana@Bhavana:~$ mkdir skill1.1
```

```
bhavana@Bhavana:~$ cd skill1.1
```

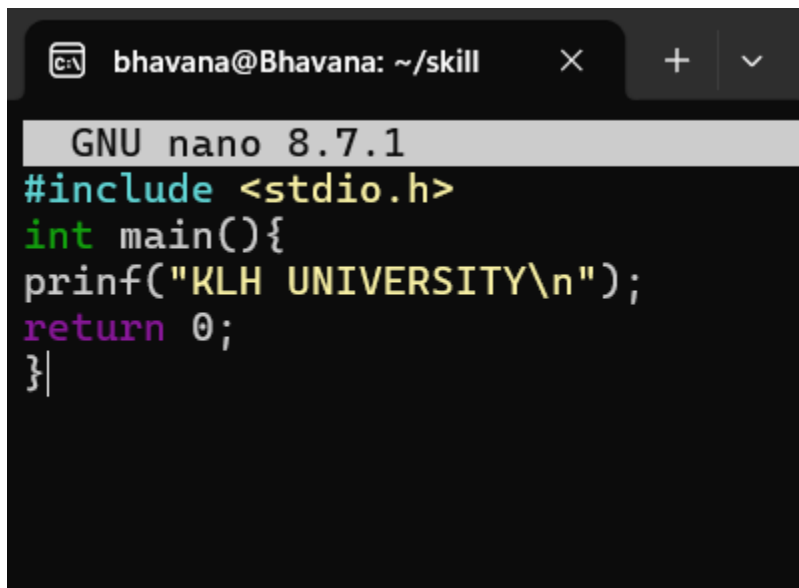
```
bhavana@Bhavana:~/skill1.1$ pwd
/home/bhavana/skill1.1
```

Step 9

```
init
bhavana@Bhavana:~/skill$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/bhavana/skill/.git/
```

Step 10

```
Initialized empty Git repository in /home/bhavana/skill/.git/
bhavana@Bhavana:~/skill$ nano main.c
bhavana@Bhavana:~/skill$
```



```
bhavana@Bhavana: ~/skill
GNU nano 8.7.1
#include <stdio.h>
int main(){
printf("KLH UNIVERSITY\n");
return 0;
}
```

Step 11

```
bhavana@Bhavana:~/skill$ nano Makefile
```

```
bhavana@Bhavana: ~/skill
GNU nano 8.7.1
CC = gcc
TARGET = bin/main
SOURCE = main.c

all:
    $(CC) $(SOURCE) -o $(TARGET)
clean:
    rm -f $(TARGET)|
```

Step 12

```
bhavana@Bhavana:~/skill$ nano Makefile
bhavana@Bhavana:~/skill$ make
Makefile:6: *** missing separator.  Stop.
bhavana@Bhavana:~/skill$ |
```

Step 13

```
Compilation terminated.
bhavana@Bhavana:~/skill$ pstree
systemd--agetty
          |--chronyd-starter--chronyd--chronyd
          |--cron
          |--dbus-daemon
          |--init-systemd(Ub
              |--SessionLeader--Relay(333)--bash--pstree
              |--init--{init}
              |--login--bash
              |--{init-systemd(Ub}
          |--networkd-dispat
          |--polkitd--3*[{polkitd}]
          |--rsyslogd--3*[{rsyslogd}]
          |--systemd--(sd-pam)
          |--systemd-journal
          |--systemd-logind
          |--systemd-resolve
          |--systemd-udev
          |--unattended-upgr--{unattended-upgr}
```

Conclusion

The Linux development environment was successfully configured with GCC, Git, project structure, and an initial Makefile. The experiment provided practical understanding of Linux shell architecture and process abstraction, including `fork()`, the `exec()` family, parent-child relationships, and process trees. System-call tracing using `strace` helped demonstrate how user-level programs interact with the Linux kernel during process creation and execution.